

MSc in Business Analytics

2025 - 2026



**“A Comparative Study of Discriminative and Generative Models for sentiment Analysis of
Employee Reviews”**

By Soraya Shirini

***Submitted to Dublin City University in partial fulfillment of the requirements for the
degree of Master’s in Business Analytics***

Table of Contents

1. Business Problem and Scope	3
2. Model Selection and Justification	3
2.1 BERT Model	3
2.2 Generative AI Model (GPT Prompting).....	4
2.3 Model Comparison Justification	4
3. Dataset Selection and Preparation	4
4. Methodology and Model Implementation	5
4.1 BERT Model Implementation	5
4.2 GPT Model Implementation	6
4.2 Constraints and Implementation Considerations	6
5. Results and Performance Analysis	6
5.1 Overview of Model Performance	6
5.2 BERT Model results	6
5.2 GPT Model results	8
5.3 Model Comparison	9
6. Ethical and Responsible AI Considerations	10
7. Business Relevance, Reflection, Limitation, and Practical Implications	11
8. Reference	12

1. Business Problem and Scope

This study focuses on sentiment analysis of employee reviews using data from Glassdoor. Many companies receive large amounts of employee feedback online, but it is difficult to read and analyze all reviews manually. As a result, important insights about employee satisfaction and workplace issues may be missed [2].

In the context of human resource analytics, organizations use employee reviews to understand job satisfaction, management quality and workplace culture. Data-driven decision-making helps organizations extract useful insights from large datasets [1]. However, most of this information is stored as unstructured text, which is difficult to process without automated methods [2]. Text analysis is valuable because it enables organizations to efficiently process large volumes of feedback and identify patterns that are not easily visible through manual analysis.

The goal of this study is to build models that can automatically classify employee reviews into sentiment categories (positive, neutral, negative). This helps organizations gain insights from large amounts of text data and supports better decision-making in areas such as employee retention and workplace improvement. For example, HR managers can use sentiment analysis dashboards to identify trends in employee dissatisfaction across departments and take targeted actions.

The model supports decision-making by identifying departments with low satisfaction, prioritizing HR interventions and providing data-driven insights to improve employee retention and workplace policies.

Therefore, sentiment analysis transforms unstructured employee feedback into actionable business insights that support strategic HR decision-making.

2. Model Selection and Justification

This study compares two types of models for sentiment classification: a discriminative model (BERT) and a generative model (GPT-based prompting).

2.1 BERT Model

BERT is a transformer-based model that processes text bidirectionally, meaning it considers both left and right context, which improves language understanding [3]. This makes BERT highly effective for text classification tasks.

BERT is a discriminative model, meaning it learns a direct relationship between input text and output labels. It is suitable for this study because sentiment classification requires predicting predefined categories using labeled data. Previous research shows that fine-tuned BERT models achieve strong performance in classification tasks such as sentiment analysis [3]. In addition, BERT benefits from transfer learning, where knowledge learned from large datasets is applied to specific tasks [3].

However, BERT has several limitations. Transformer-based models require significant computational resources for training and deployment [4]. BERT must also be fine-tuned for each

new task, which reduces flexibility. Furthermore, its fixed input length may lead to information loss when long texts are truncated [3].

In this study, BERT is used through fine-tuning on labeled sentiment data.

2.2 Generative AI Model (GPT Prompting)

A generative AI model is used through prompt-based classification. Unlike BERT, GPT is a generative model, meaning it produces text outputs rather than directly predicting labels. Generative models such as GPT can perform tasks using zero-shot or few-shot learning without additional training [5].

However, generative models have several limitations. They may produce inconsistent outputs and are sensitive to prompt design [5]. In addition, they can generate incorrect or misleading information, known as hallucination [6]. These models may also reflect biases present in training data, which can affect fairness and reliability [7].

In this study, GPT is used through prompt-based classification without additional training.

2.3 Model Comparison Justification

BERT is suitable for structured classification tasks with labeled data, while GPT provides flexibility through prompt-based reasoning. BERT is expected to achieve higher accuracy due to task-specific training, whereas GPT offers adaptability without requiring labeled data. This comparison allows evaluation of accuracy versus flexibility in a real-world business context.

3. Dataset Selection and Preparation

This study uses a publicly available dataset of employee reviews from Glassdoor. Glassdoor allows employees to anonymously share feedback about their workplace, including ratings and comments. This type of data is highly relevant for organizations that want to monitor employee satisfaction, detect potential workplace issues and improve internal management strategies. This dataset is appropriate for the business problem because it reflects real-world employee feedback used in HR analytics, making the findings directly applicable to organizational decision-making.

The dataset contains 838,566 reviews from multiple firms providing a large and diverse sample suitable for training machine learning models. For this study, only reviews from London, United Kingdom were selected to ensure consistency in workplace. The large size of the dataset helps improve model performance and generalization.

Table 1: Dataset Structure

Field	Description
firm	Company name
Overall_rating	Employee rating
headline	Review summary
pros	Positive comments
cons	Negative comments

For the analysis, the textual fields were combined into a single variable called review, which represent the full employee comment. To create labels for the classification task, the original 1-5 rating scale was converted into three sentiment categories.

Table 2: Sentiment Label Mapping and Distribution (London subset)

Rating	Sentiment	Review Counts	Percentage
4-5	Positive	35328	60%
3	Neutral	13139	22%
1-2	Negative	10198	17%

The dataset shows class imbalance, with a higher proportion of positive reviews, which is important to consider during model evaluation.

The dataset was divided into training and test sets using 80:20 split. Stratified sampling was used to maintain the same sentiment distribution in both datasets. The training set is used to train the model, while the test set is used to evaluate model performance on unseen data.

The following preprocessing steps were applied to ensure that the text data is in a suitable format for model training.

- Missing values replaced with empty strings
- Headline, pros, cons combined into single review text field
- Text tokenized using the WordPiece tokenizer from bert-base-uncased
- Text truncated to meet BERT input requirements
- Basic text cleaning applied (e.g., removing extra spaces)

4. Methodology and Model Implementation

This study applies two approaches to sentiment classification. A fine-tuned BERT model and a prompt-based GPT model. Both models are evaluated using the same dataset and performance metrics to ensure a fair comparison.

4.1 BERT Model Implementation

The BERT model was implemented using the Hugging Face Transformers library. Text data was tokenized using the wordPiece tokenizaer from bert-base-uncased model. The dataset was split into training and test sets using an 80:20 ratio with stratified sampling to preserve class distribution.

The model was fine-tuned on labeled sentiment data using supervised learning. Training was performed using the Hugging Face Trainer with key hyperparameters including a learning rate of 2e-5, batch size of 8, and one training epoch due to computational

constraints. The model was trained to predict three sentiment classes (negative, neutral, positive).

After training, the model was evaluated on the test dataset using accuracy, precision, recall, and F1-score, along with confusion matrix analysis to assess classification errors.

4.2 GPT Model Implementation

The GPT model was applied using prompt-based classification without additional training. Each employee review was passed to a locally hosted generative model (via Ollama) using a structured prompt.

The prompt design follows a zero-shot approach and explicitly instructs the model to classify each review into one of one of three labels (negative, neutral, positive) and return only a single-word output.

Due to computational limitations and slower inference speed, GPT was evaluated on a stratified subset of 10,000 samples from the test dataset while maintaining the same class distribution.

The model output was mapped to numerical labels for comparison with true labels. The same evaluation metrics used for BERT (accuracy, precision, recall, and F1-score) were applied to ensure consistency.

4.2 Constraints and Implementation Considerations

This study is limited by computational resources, which restrict the number of training epochs for BERT and require sampling for GPT evaluation. In addition, GPT performance depends on prompt design, which may affect consistency and reproducibility.

5. Results and Performance Analysis

5.1 Overview of Model Performance

This study evaluates the performance of two models for sentiment classification: a fine-tuned BERT model and a generative AI model (GPT-based prompting). The models are compared using standard classification metrics including accuracy, recall, precision and F1-score.

5.2 BERT Model results

Table 3 : BERT Classification Performance

Class	Precision	Recall	F1-score
Negative (0)	0.78	0.66	0.71
Neutral (1)	0.52	0.44	0.48
Positive (2)	0.82	0.91	0.87

Metric	Precision
Accuracy	0.76
Macro Avg F1	0.68
Weighted Avg F1	0.75

The BERT model achieved an overall accuracy of 0.76. the model performed best on the positive class, achieving an F1-score of 0.87 and a recall of 0.91. This suggests that positive reviews are easier to identify due to clearer and more consistent language patterns. Performance on negative class was moderate, with an F1-score of 0.71. While the model was able to detect negative sentiment, some misclassification occurred. The weakest performance was observed in the neutral class, with an F1-score of 0.48. This indicates that the model struggles to distinguish neutral reviews, likely due to the ambiguity of neutral language and overlap with other sentiment categories.

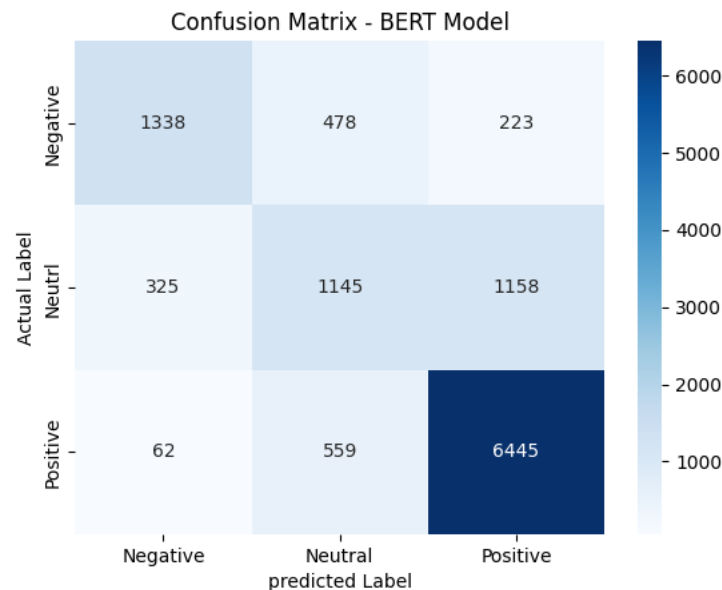


Figure 1:

Figure 1 presents the confusion matrix for the BERT model. The result confirms that the model performs strongly on the positive class, with most positive reviews correctly classified. In contrast, the neutral class shows the highest level of misclassification,

particularly being predicted as positive. This indicates that the model struggles to distinguish neutral sentiment due to overlap with other classes. Some negative reviews are also misclassified as neutral, suggesting difficulty in identifying weaker negative expressions.

5.2 GPT Model results

To evaluate the GPT-based model, a stratified subset of 10000 samples from the test dataset was used. This subset preserves the original class distribution and allows for a fair comparison with BERT model.

Table4: GPT Classification Performance

Class	Precision	Recall	F1-score
Negative (0)	0.42	0.88	0.57
Neutral (1)	0.25	0.35	0.29
Positive (2)	0.91	0.49	0.64

Metric	Score
Accuracy	0.53
Macro Avg F1	0.50
Weighted Avg F1	0.55

The GPT-based model achieved an overall accuracy 0.53, and a weighted F1-score of 0.55, indicating moderate performance in sentiment classification.

The model performed well in terms of precision for the positive class (0.91), suggesting that when GPT predicts a review as positive, it is usually correct. However, the recall is relatively low(0.49), meaning that many positive reviews are not correctly identified.

For the negative class, the model achieves a high recall of 0.88, indicating the most negative reviews are detected. However, the low precision (0.42) shows that the model tends to over-predict negative sentiment, leading to incorrect classification.

The neutral class shows the weakest performance, with an F1-score of 0.29, indicating difficulty in handling ambiguous sentiment.

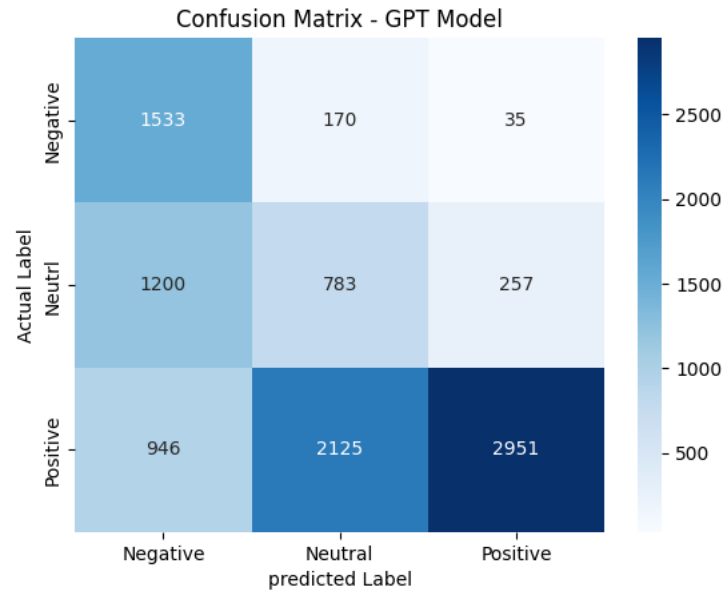


Figure 2: GPT Confusion Matrix

Figure 2 shows the confusion matrix for the GPT model. The results indicate that the model frequently misclassifies positive reviews as neutral or negative. In addition, neutral reviews are often confused with both positive and negative classes. This suggests that the GPT model lacks consistency in classification and struggles to clearly separate sentiments categories.

5.3 Model Comparison

Table 5: Model Comparison Summary

Metric	BERT	GPT
Accuracy	0.76	0.53
Weighted F1	0.75	0.55
Negative F1	0.71	0.57
Neutral F1	0.48	0.29
Positive F1	0.87	0.64

The result show that the BERT model significantly outperforms the GPT-based across all evaluation metrics. BERT perform particularly well in identifying positive sentiment. Both models struggle with neutral class; however, GPT perform considerably worse, highlighting its difficulty in handling ambiguous text.

Overall, the result indicate that BERT provide more stable and accurate prediction, while GPT shows lower consistency due to its prompt-based nature.

6. Ethical and Responsible AI Considerations

Artificial Intelligence systems are increasingly used to support decision-making in organizations, including human resource analytics. However, these systems involve risk, uncertainty and potential harm, making ethical considerations essential. Trust in AI should not be automatic but must be earned through responsible design and governance. In this study, the use of sentiment analysis on employee reviews raises several ethical concerns aligned with the principles of trustworthy AI.

The first key issue is fairness. AI models such as BERT and GPT learn from historical data, which may contain biases. For example, employee reviews may reflect biased opinions about management, workplace culture, or specific groups. If these biases are learned by the model, the system may produce unfair or misleading classifications. This can result in biased decisions, such as incorrectly identifying workplace problems or misinterpreting employee sentiment, ultimately reducing trust in decision-making systems.

Transparency and explainability are important concerns in this study. Models such as BERT are often considered “black-box” systems, making it difficult to understand how predictions are generated. In an HR context, this could mean that a negative sentiment label is assigned to an employee review without a clear reason. As a result, HR managers may misinterpret employee concerns or take incorrect actions, such as prioritizing the wrong workplace issues. This lack of explainability can reduce trust if employees feel that decisions are based on unclear or unexplainable systems.

Accountability is also critical. AI systems cannot take responsibility for their decisions, instead, organizations must ensure that outputs are monitored, evaluated and corrected when necessary. For example, if the model incorrectly classifies positive reviews as negative, management may believe there is a serious workplace issue when there is not. If no one is responsible for checking or correcting, decisions may be based on flawed insights, leading to poor management strategies and potential reputational damage.

Privacy and data governance must also be considered. Employee reviews may still contain sensitive information about managers or internal issues. If organizations use this data without proper safeguards, employees may feel their feedback is being monitored or misused, which can reduce trust and discourage honest reviews in the future.

Human agency and oversight are essential to ensure responsible use. The model should support decision-making rather than replace it. For example, if HR teams rely entirely on automated sentiment scores to evaluate employee satisfaction, they may ignore important context or nuanced feedback. This can lead to oversimplified decisions, such as implementing unnecessary policies or missing deeper workplace problems.

Technical robustness and reliability are equally important. AI models may produce inconsistent outputs, especially when new types of reviews are introduced. For instance, if the model misclassifies sarcastic or mixed reviews, HR teams may receive inaccurate insights about employee sentiments. Over time, inconsistent results can reduce confidence in the system and limit its usefulness in decision-making.

Overall, these ethical issues directly impact business decision-making, organizational trust and compliance. If AI systems are perceived as biased, unclear, or unreliable, employees may lose trust in management decisions, and organizations may face reputational or regulatory risks. Therefore, ensuring fairness, transparency, accountability, and human oversight is essential for responsible use of AI in HR analytics.

7. Business Relevance, Reflection, Limitation, and Practical Implications

The results show that the BERT model is more suitable for real-world business use than the GPT-based model. BERT showed better performance, making it more reliable for tasks such as analyzing employee feedback. In contrast, GPT showed lower accuracy which may reduce trust in business decisions.

In practice, accuracy and consistency are more important than flexibility for this use case. Organizations need reliable insights to support decisions related to employee satisfaction, retention and workplace improvement. BERT can process large volumes of data quickly after training, making it more scalable and cost-efficient. GPT prompting is slower and more expensive.

The BERT model can be integrated into HR analytics workflows to automatically analyze employee reviews. For example, it can help track sentiment trends, identify common issues, and support data-driven HR strategies. However, it should be used as a decision-support tool with human oversight, rather than replacing human judgment.

This study has several limitations. The dataset is imbalanced, which may influence model performance. In addition, using only three sentiments may oversimplify complex feedback.

With more time improvement could include using larger and more balanced dataset, improving prompt design and exploring methods to better classify neutral sentiment.

Overall, this project shows that applying AI in business requires not only technical performance but also consideration of reliability, cost, scalability and practical use.

8. Reference

[1]F. Provost and T. Fawcett, “Chapter 2. Business Problems and Data Science Solutions,” in Data Science for Business, United States: O’Reilly Media, Incorporated, 2013.

[2]J. A. Morgan and D. S. Chapman, “Examining the characteristics and effectiveness of online employee reviews,” Computers in human behavior reports, vol. 16, Art. no. 100471, Dec. 2024, doi: 10.1016/j.chbr.2024.100471. Available at:
https://dcu.primo.exlibrisgroup.com/permalink/353DCU_INST/clbqq1/cdi_crossref_citationtrail_10_1016_j_chbr_2024_100471

[3]Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” Proceedings of the 2019 Conference of the North, vol. 2019, pp. 4171–4186, 2019, doi: 10.18653/v1/n19-1423. Available at:
https://dcu.primo.exlibrisgroup.com/permalink/353DCU_INST/clbqq1/cdi_jndl_porta_oai_ndlsearch_ndl_go_jp_R100000136_I1361699993573121280

[4]E. Strubell, A. Ganesh, and A. McCallum, “Energy and Policy Considerations for Deep Learning in NLP,” June 2019, doi: 10.48550/arxiv.1906.02243. Available at:
https://dcu.primo.exlibrisgroup.com/permalink/353DCU_INST/clbqq1/cdi_arxiv_primary_1906_02243

[5]P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, “Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing,” ACM Computing Surveys, vol. 55, no. 9, Art. no. 195, Sept. 2023, doi: 10.1145/3560815.

https://dcu.primo.exlibrisgroup.com/permalink/353DCU_INST/clbqq1/cdi_crossref_primary_101145_3560815

[6]Z. Ji et al., “Survey of Hallucination in Natural Language Generation,” ACM Computing Surveys, vol. 55, no. 12, Art. no. 248, Dec. 2023, doi: 10.1145/3571730.

https://dcu.primo.exlibrisgroup.com/permalink/353DCU_INST/clbqq1/cdi_crossref_citationtrail_10_1145_3571730

[7]I. O. Gallegos et al., “Bias and Fairness in Large Language Models: A Survey,” Computational Linguistics, vol. 50, no. 3, pp. 1097–1179, Sept. 2024, doi: 10.1162/coli_a_00524.

https://dcu.primo.exlibrisgroup.com/permalink/353DCU_INST/clbqq1/cdi_unpaywall_primary_10_1162_coli_a_00524